Learning Experience Questionnaire

Conventional Instruction Version

Participant ID: _____

Section A. Perceived Learning Support (PLS)

PLS1. The classroom instruction helped me better understand the mechanisms of traffic conflict formation in ramp merging scenarios.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PLS2. The classroom instruction improved my understanding of vehicle interactions in weaving sections.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PLS3. The classroom case analysis and teaching materials helped me connect theoretical knowledge with real-world traffic scenarios.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PLS4. The classroom instruction made complex traffic processes easier to understand.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PLS5. The post-instruction review and classroom discussion helped me more clearly identify the causes of traffic conflicts.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

Section B. Cognitive Engagement (CE)

CE1. I was able to maintain a high level of focus during the classroom learning process.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

CE2. The classroom case-analysis tasks encouraged me to actively think about how traffic conflicts are formed.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

CE3. I was more willing to participate in discussions and analysis during the classroom session.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

CE4. The classroom teaching content increased my interest in learning traffic conflict-related topics.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

CE5. During classroom case analysis, I was willing to invest more effort in understanding vehicle interactions.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

Section C. Perceived Realism and Interactivity (PRI)

PRI1. The ramp merging and weaving cases used in the classroom had clear contextual representation and supported my learning.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PRI2. The classroom presentation of dynamic vehicle interactions helped me understand how traffic conflicts are formed.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PRI3. The classroom diagrams, cases, and discussion activities provided a stronger sense of involvement in traffic scenarios.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PRI4. The classroom instruction helped me observe and understand traffic processes that are difficult to grasp through text-based explanation alone.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PRI5. The classroom case analysis and discussion made the learning process more intuitive.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

Section D. Learning Satisfaction and Behavioral Intention (LSBI)

LSBI1. I am satisfied with the classroom teaching approach used in this course.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

LSBI2. I believe this teaching approach has practical value for transportation engineering courses.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

LSBI3. I would like to have similar classroom case-analysis activities in other transportation-related courses.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

LSBI4. The classroom instruction improved my overall learning experience in this topic.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

LSBI5. I would recommend this teaching approach to other students.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree